

VQ-VAE a posteriori with Geodesic Quantization

Deep Learning and Applied AI, Sapienza Università di Roma – 2025

Abstract

In this project we study an alternative to the classical VQ-VAE pipeline: instead of learning a discrete latent space jointly with the encoder, we first train a standard Variational Autoencoder (VAE) on the MNIST dataset and perform vector quantization ‘a posteriori’. To better capture the intrinsic geometry of the latent space, quantization is performed using geodesic distances – computed over a k-NN graph via Landmark Isomap, which makes the approach tractable on the full 60 000-image training set – instead of plain Euclidean distances.

We compare this approach against a convolutional VQ-VAE trained end-to-end with spatial vector quantization and, to isolate the effect of the distance metric itself, against an ablation that applies the same clustering pipeline with Euclidean distances on the same latents. An autoregressive GRU prior is trained over the resulting discrete codes to generate new samples.

Our results show that geodesic quantization yields consistently better-structured discrete representations than its Euclidean counterpart: clustering purity and normalized mutual information against the true digit labels are both higher under the geodesic metric, indicating that respecting the latent manifold’s geometry produces more semantically coherent codes.

1. Introduction

Unsupervised learning is a fundamental tool for discovering structured representations in complex data, without the need for explicit labels. In this category, we can find Variational Autoencoders (VAE) and their discrete variants, such as Vector Quantized Variational Autoencoders (VQ-VAE), and they are really effective in tasks such as compression or sample generation. However, classical VQ-VAE learns the code book jointly with the encoder and relies on euclidean distances, which may ignore the manifold’s intrinsic geometry.

In this work, we propose a modular, geometry-aware pipeline:

1. Train a continuous VAE on MNIST.
2. Build a k-NN graph ($k = 10$) over the learned latent codes and compute geodesic distances.
3. Apply geodesic k-means to obtain a discrete code book.
4. Train an auto regressive RNN on the discrete codes to generate new samples.

By doing all of this, we want to capture better the structure of the latent space before applying quantization, so we will obtain more coherent and useful discrete representations. Finally, we compare the performance of the alternative that we implemented with the classical VQ-VAE.

2. Related Work

Several works have studied non-Euclidean distances in latent spaces. Tenenbaum et al. [4] introduced Isomap, which approximates geodesics via a k-NN graph to recover manifold geometry. Asgharbeygi and Maleki [1] proposed geodesic k-means by replacing Euclidean with shortest-path distances

on a graph. Yang et al. [7] demonstrated that geodesic clustering in VAEs yields high-quality clusters despite latent distortions.

2.1. Our contribution

Unlike standard VQ-VAE, we first train a continuous VAE and then perform a posteriori geodesic quantization via a k-NN graph and geodesic k-means, followed by autoregressive modeling of the resulting discrete codes. This geometry-aware, decoupled approach aims to improve latent code coherence and enable generation over the code book.

3. Method

3.1. Baseline: End-to-End VQ-VAE

Our baseline is a convolutional Vector Quantized Variational Autoencoder (VQ-VAE) [5]. Unlike a design that collapses each image into a single latent vector, the encoder here produces a *spatial grid* of latent vectors (7×7 positions, each of dimension $D = 16$), and each position is quantized independently against a shared code book of size $K = 64$. With a 7×7 grid, the number of representable discrete configurations per image is K^{49} , rather than K .

The code book is updated via an exponential moving average (EMA) of the encoder outputs assigned to each entry [3], rather than by gradient descent on a code book loss term. Denoting by $n_i^{(t)}$ the number of encoder vectors assigned to code i at training step t and by $\sum z_e^{(t)}(x)$ the sum of those vectors,

$$N_i^{(t)} = \gamma N_i^{(t-1)} + (1 - \gamma) n_i^{(t)} \quad (1)$$

$$m_i^{(t)} = \gamma m_i^{(t-1)} + (1 - \gamma) \sum_{z_e(x) \rightarrow i} z_e^{(t)}(x) \quad (2)$$

$$e_i^{(t)} = m_i^{(t)} / N_i^{(t)} \quad (3)$$

with decay $\gamma = 0.99$. The model is then trained end-to-end with

$$\mathcal{L}_{\text{VQ-VAE}} = \underbrace{\text{BCE}(x, \hat{x})}_{\text{reconstruction}} + \underbrace{\beta \|\text{sg}[z_e(x)] - e\|^2}_{\text{commitment}} \quad (4)$$

where $z_e(x)$ is the encoder output at a given spatial position, e the assigned code book entry, $\text{sg}[\cdot]$ the stop-gradient operator, and β the commitment weight. Since the code book is updated by EMA rather than by gradient, the loss no longer needs a separate code book term, unlike the standard formulation.

3.2. Proposed Method and Contribution: ‘a posteriori’ Geodesic Quantization

In contrast, our method decouples quantization from VAE training and exploits manifold geometry:

1. Train continuous VAE.

- Convolutional encoder/decoder; the encoder output is flattened and projected to a vector latent of dimension $D = 20$ (mean and log-variance).
- Loss: ELBO = BCE + $\text{KL}(q_\phi(z|x) \| p(z))$.
- Optimizer: Adam, lr = 10^{-3} , batch 128, epochs 20.

2. Construct k-NN graph and embed geodesic distances.

- Extract latents $\{z_i\} \subset \mathbb{R}^D$ for the full training set (60 000 points).
- Build a k-NN graph with $k = 10$ nearest neighbors.
- Computing the full $N \times N$ geodesic distance matrix is infeasible at this scale (~ 14 GB for $N = 60\,000$). Instead, we use *Landmark Isomap* [2]: geodesic distances are computed via Dijkstra’s algorithm from a set of $L = 500$ landmark points to all N points, and all points are then embedded into a 10-dimensional space via Landmark MDS.

3. Apply geodesic k-means clustering.

- Run K-means with $K = 64$ clusters on the Landmark Isomap embedding of the full training set.
- Cluster centroids $\{c_j\}_{j=1}^K$ are computed in the *original* latent space (mean of the latent vectors assigned to each cluster), so that they remain decodable by the VAE decoder.
- As a control, the same K-means pipeline is run with Euclidean distance directly on the latents (same K , same data), to isolate the effect of the distance metric from other factors.

4. Auto regressive modeling.

- A GRU-based RNN is trained to predict each code of a VQ-VAE code grid from the previous codes of that *same* image, in raster order (sequence length 49, i.e. the 7×7 grid flattened).

- Embedding dim 64, hidden dim 128, batch 256, epochs 15.

Our main contribution is the ‘a posteriori’ geodesic quantization pipeline (steps 2–3), which leverages the intrinsic latent manifold to produce more coherent discrete codes, validated against a matched Euclidean baseline, and the raster-order autoregressive prior (step 4), which models the dependencies between codes within a single image.

4. Experiments and results

4.1. Experimental setup

All experiments were conducted on the MNIST dataset (60 000 train / 10 000 test, 28×28 grayscale). We trained the following:

• Continuous VAE

- Convolutional encoder: $28 \times 28 \times 1 \rightarrow 14 \times 14 \times 32 \rightarrow 7 \times 7 \times 64$, flattened and projected to $\mu, \log \sigma^2 \in \mathbb{R}^{20}$; decoder mirrors this structure with transposed convolutions.
- Latent dimension $D = 20$.
- Loss: ELBO = BCE + KL divergence.
- Optimizer: Adam with lr = 1×10^{-3} .
- Batch size: 256; Epochs: 20.

• Baseline VQ-VAE

- Convolutional encoder/decoder, spatial code grid (7×7 positions, embedding dim 16).
- Code book size $K = 64$, updated via EMA (decay $\gamma = 0.99$).
- Commitment weight $\beta = 0.25$.
- Loss: BCE + commitment (Eq. 4).
- Optimizer: Adam, lr = 1×10^{-3} .
- Batch size: 256; Epochs: 20.

• Geodesic quantization

- Extract 60 000 latent vectors from the trained VAE.
- Build a k-NN graph with $k = 10$.
- Compute geodesic distances via Landmark Isomap: Dijkstra from $L = 500$ landmarks, embedding into 10 dimensions via Landmark MDS.
- Run K-means with $K = 64$ clusters on the embedding; centroids are mapped back to the original latent space.
- *Euclidean control*: K-means with $K = 64$ run directly on the same latents, without the geodesic embedding step.

• Autoregressive RNN

- GRU model: embedding dim 64, hidden dim 128.

- Sequence length: 49 tokens (the 7×7 VQ-VAE code grid of a single image, flattened in raster order); vocabulary size: 64.
- Loss: cross-entropy (NLL).
- Optimizer: Adam, $\text{lr} = 1 \times 10^{-3}$.
- Batch size: 256; Epochs: 15.

For the Geo-VQ model, the pre-trained VAE encoder was used to generate latent vectors for the entire training set. A k -NN graph ($k = 10$) was built over these latents, and geodesic distances were approximated via Landmark Isomap (Dijkstra from 500 landmarks, followed by Landmark MDS), which avoids the $O(N^2)$ memory cost of computing the full pairwise geodesic distance matrix. K-means with $K = 64$ was then applied to the resulting embedding to obtain the quantization code book, and the same procedure with plain Euclidean distances on the original latents was run as a control to isolate the contribution of the geodesic metric itself.

4.2. Metrics

To compare the performance of the models we used the following metrics:

BCE Binary cross-entropy over all pixels and test samples:

$$\text{BCE} = - \sum_{i=1}^N \sum_{j=1}^{28 \times 28} [x_{i,j} \ln \hat{x}_{i,j} + (1 - x_{i,j}) \ln(1 - \hat{x}_{i,j})]. \quad (5)$$

Regularizer For VAE: KL divergence $\text{KL}(q_\phi(z|x) \| p(z))$; for VQ-VAE: commitment loss $\beta \| \text{sg}[z_e(x)] - e \|^2$.

Total Loss VAE: ELBO = BCE+KL. VQ-VAE: BCE+commitment.

NLL (AR) Sequence negative log-likelihood per token:

$$\text{NLL} = - \frac{1}{M} \sum_{t=1}^M \ln P(c_t | c_{<t}). \quad (6)$$

4.3. Quantitative comparison

Table 1: Reconstruction quality on the test set. Terms are reported separately rather than summed into a single loss (see Section 4.2).

Model	BCE/img	SSIM	PSNR	Other
VAE	75.6	0.872	19.8 dB	KL = 25.7
VQ-VAE	95.8	0.826	17.6 dB	Perplexity = 26.2/64
Geo-VQ	187.2	0.603	13.8 dB	—
Eucl-VQ (control)	186.4	0.603	13.8 dB	—

Geo-VQ and VQ-VAE are not directly comparable in reconstruction quality: Geo-VQ quantizes a single global vector (64 possible reconstructions in total across the whole test set), whereas VQ-VAE quantizes a 7×7 spatial grid (64^{49} combinations). This is an inherent trade-off of the ‘a posteriori’ design rather than a shortcoming of the geodesic metric itself; consequently, the relevant comparison for the

Table 2: Clustering quality of the a posteriori quantization, geodesic vs. Euclidean, on the same VAE latents with the same $K = 64$, evaluated against the true digit label.

Method	Purity	NMI
Geodesic	0.953	0.651
Euclidean	0.926	0.611

geodesic quantization method is the ablation in Table 2, not raw reconstruction quality against VQ-VAE.

Table 2 isolates the effect of the distance metric: with the same latents and the same number of clusters, geodesic quantization produces clusters that align more closely with the true digit classes than Euclidean quantization, on both purity and NMI. The autoregressive prior trained on the VQ-VAE code grids reaches a final NLL of 0.886 nats/token, indicating that the codes within an image carry substantial predictable structure that the GRU can exploit to generate new, coherent grids (Section 4.4).

4.4. Visualizations (qualitative comparison)

Now, we see the visualizations that have been generated:

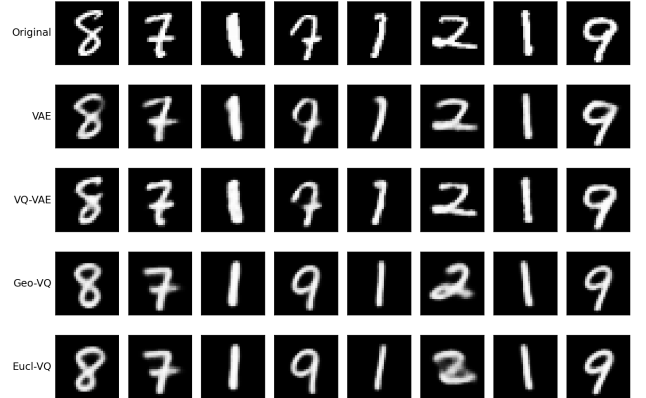


Figure 1: Reconstructions of test images. Rows, top to bottom: original, VAE, VQ-VAE, Geo-VQ, Eucl-VQ (control).

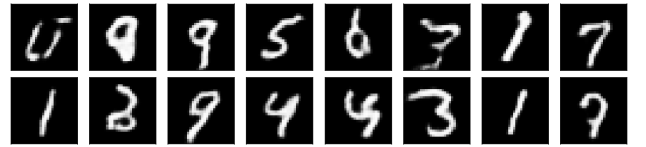


Figure 2: 16 digits generated from scratch by the autoregressive GRU prior over VQ-VAE code grids, decoded through the VQ-VAE decoder.

From Figure 1, we can extract the following conclusions:

1. **VAE:** reconstructions closely follow the input digits, with the mild blurriness characteristic of a Gaussian latent space and a pixel-wise reconstruction loss.
2. **VQ-VAE:** reconstructions are sharp and every digit remains identifiable. A mild striping pattern is visible in some examples, a known artifact of the transposed

convolutions used in the decoder; it does not affect legibility.

3. **Geo-VQ and Eucl-VQ:** both quantize a single latent vector per image into one of 64 possible codes, so reconstructions are visibly smoother and closer to a class prototype than to the specific input digit. The two are visually similar to each other, as expected – the meaningful difference between the geodesic and Euclidean quantizations lies in how well their clusters align with the true digit classes (Table 2), not in reconstruction sharpness, since both decode through the same 64-centroid bottleneck.

Figure 2 shows digits sampled from scratch by the autoregressive prior, without conditioning on any real image: the GRU generates a full 7×7 code grid token by token, which is then decoded by the VQ-VAE. Most samples are recognizable as specific digits (e.g. 7, 1, 6, 9, 0) with plausible stroke variation, indicating that the raster-order prior captures meaningful local dependencies between neighboring codes rather than producing incoherent grids. The same mild striping artifact observed in VQ-VAE reconstructions is visible here too, since generation reuses the same decoder.

5. Conclusions

Our experiments on MNIST show that quantizing a VAE’s latent space with geodesic distances produces discrete representations that align more closely with the underlying data classes than an equivalent Euclidean quantization: with the same latents and the same code book size, geodesic k-means achieves both higher clustering purity and higher NMI against the true digit labels (Table 2). This supports the hypothesis that respecting the geometry of the latent manifold, rather than treating it as locally Euclidean, yields a more semantically meaningful discretization.

Separately, a convolutional VQ-VAE with spatial vector quantization produces sharp, legible reconstructions across all ten digit classes, and an autoregressive GRU prior trained on its code grids reaches a low per-token NLL (0.886 nats) and generates new, recognizable digits from scratch – indicating that the codes within an image carry learnable local structure that a simple sequence model can exploit.

The ‘a posteriori’ geodesic quantization, by design, quantizes a single latent vector per image and is therefore not directly comparable to the spatial VQ-VAE in reconstruction fidelity; its contribution is best read through the clustering ablation rather than through raw pixel-level metrics. Remaining limitations – a mild striping artifact in VQ-VAE reconstructions and a commitment loss that grows steadily over training – point to natural next steps, such as replacing the transposed convolutions with resize-and-convolve upsampling and normalizing encoder outputs before quantization.

References

- [1] Asgharbeygi, N. and Maleki, A. Geodesic clustering in deep generative models. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 30(10):1732–1743, 2008. <https://arxiv.org/abs/0809.2251>.
- [2] de Silva, V. and Tenenbaum, J. B. Global versus local methods in nonlinear dimensionality reduction. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2002.
- [3] Razavi, A., van den Oord, A., and Vinyals, O. Generating diverse high-fidelity images with VQ-VAE-2. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019. <https://arxiv.org/abs/1906.00446>.
- [4] Tenenbaum, J. B., De Silva, V., and Langford, J. C. A global geometric framework for nonlinear dimensionality reduction. *Science*, 290(5500):2319–2323, 2000. <https://science.sciencemag.org/content/290/5500/2319>.
- [5] van den Oord, A., Vinyals, O., and Kavukcuoglu, K. Neural discrete representation learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017. <https://arxiv.org/abs/1711.00937>.
- [6] Wang, Z., Bovik, A. C., Sheikh, H. R., and Simoncelli, E. P. Image quality assessment: from error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, 2004.
- [7] Yang, G., Hypki, A., and Welling, M. Geodesic clustering in deep generative models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018. https://papers.nips.cc/paper_files/paper/2018/file/0c576f7c923121e69edce3aa3734fffa-Paper.pdf.